

Gumbel-max Watermark

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Lemma 1.

If $X \sim U(0, 1)$, $Y = -\ln(X)$, then $Y \sim \text{Exp}(1)$

Proof.

This is clear by manipulating their CDF.

Now

$$\arg \max_w \frac{\log U_w}{P_w} \iff \arg \min_w \frac{-\log U_w}{P_w} \iff \arg \min_w \frac{\text{Exp}(1)}{P_w}$$

Note (by property of Exp or Gamma, as we learned in stat 450),

$$Y_w := \frac{\text{Exp}(1)}{P_w} \sim \text{Exp}(\lambda = P_w)$$

By stat 333,

Lemma 2.

Let $\{X_i\}_{i=1}^n$ be a sequence of independent rvs where $X_i \sim \text{EXP}(\lambda_i)$, $i = 1, 2, \dots, n$. For $j \in \{1, 2, \dots, n\}$, $\mathbb{P}(X_j = \min\{X_1, X_2, \dots, X_n\}) = \frac{\lambda_j}{\sum_{i=1}^n \lambda_i}$

Proof.

$$\begin{aligned}
& \mathbb{P}(X_j = \min\{X_1, X_2, \dots, X_n\}) \\
&= \mathbb{P}(X_j < X_1, X_j < X_2, \dots, X_j < X_n) \\
&= \int_0^\infty \mathbb{P}(X_j < X_1, X_j < X_2, \dots, X_j < X_n \mid X_j = x) \lambda_j e^{-\lambda_j x} dx \\
&= \int_0^\infty \mathbb{P}(X_1 > x, X_2 > x, \dots, X_n > x) \lambda_j e^{-\lambda_j x} dx \quad (\text{since } X_j \text{ is independent of } \{X_i\}_{i \neq j}) \\
&= \int_0^\infty \mathbb{P}(X_1 > x) \mathbb{P}(X_2 > x) \cdots \mathbb{P}(X_n > x) \lambda_j e^{-\lambda_j x} dx \\
&= \int_0^\infty e^{-\lambda_1 x} e^{-\lambda_2 x} \cdots e^{-\lambda_n x} \lambda_j e^{-\lambda_j x} dx \\
&= \int_0^\infty \lambda_j e^{-(\sum_{i=1}^n \lambda_i)x} dx \\
&= \frac{\lambda_j}{\sum_{i=1}^n \lambda_i} \int_0^\infty \left(\sum_{i=1}^n \lambda_i \right) e^{-(\sum_{i=1}^n \lambda_i)x} dx \\
&= \frac{\lambda_j}{\sum_{i=1}^n \lambda_i}
\end{aligned}$$

□

Therefore

$$\mathbb{P}(Y_w \text{ is the minimum}) = \frac{P_w}{1} = P_w$$

and $\{\arg \min_{k \in W} Y_k = w\} = \{Y_w \text{ is minimum}\}$ as events, so we are done.

Proof of Theorem 2.1

Consider equation (30):

- Consider the 1st inequality.

Observe

$$\mathbb{E}_1[T_h(Y_{1:n})] = 1 \cdot \mathbb{P}_1(T_h = 1) + 0 \cdot \mathbb{P}_1(T_h = 0) = \mathbb{P}_1(T_h = 1)$$

And so

$$1 - \mathbb{E}_1[T_h(Y_{1:n})] = \mathbb{P}_1(T_h = 0)$$

Write

$$\mathbb{P}_1(T_h = 0) = \mathbb{P}_1\left(\sum_{t=1}^n h(Y_t) < \gamma_{n,\alpha}\right) = \mathbb{P}_1\left(\sum_{t=1}^n -h(Y_t) > -\gamma_{n,\alpha}\right)$$

If $\sum_{i=1}^n -h(Y_t) > -\gamma_{n,\alpha}$, then $\theta \sum_{i=1}^n -h(Y_t) > -\theta \gamma_{n,\alpha}$, thus

$$\exp\left(\sum_{i=1}^n -\theta h(Y_t)\right) > \exp(-\theta \gamma_{n,\alpha})$$

By Markov, for nonnegative Z , $a > 0$, $\mathbb{P}(Z \geq a) \leq \frac{\mathbb{E}[Z]}{a}$, so

$$\mathbb{P}_1 \left(\sum_{i=1}^n -h(Y_t) > -\gamma_{n,\alpha} \right) \leq \frac{\mathbb{E}_1[\exp(\sum_{i=1}^n -\theta h(Y_t))]}{\exp(-\theta \gamma_{n,\alpha})}$$

(In discrete case, note $\mathbb{P}(Z > a) \leq \mathbb{P}(Z \geq a)$. In cts case, they are equal.)

- Consider the 2nd inequality

Since

$$\mathbb{E}_{H_1} \exp \left(\sum_{t=1}^n -\theta h(Y_t) \right) = \mathbb{E}_{H_1} \left[\prod_{t=1}^n \exp(-\theta h(Y_t)) \right]$$

By tower rule,

$$\begin{aligned} \mathbb{E}_{H_1} \left[\prod_{t=1}^n \exp(-\theta h(Y_t)) \right] &= \mathbb{E}_{H_1} \left[\mathbb{E}_{H_1} \left[\prod_{t=1}^n \exp(-\theta h(Y_t)) \mid (w_{1:(t-1)}, \zeta_{1:(t-1)}) \right] \right] \\ &= \mathbb{E}_{H_1} \left[\left(\prod_{t=1}^{n-1} \exp(-\theta h(Y_t)) \right) \mathbb{E}_{H_1} [\exp(-\theta h(Y_n)) \mid (w_{1:(t-1)}, \zeta_{1:(t-1)})] \right] \end{aligned}$$

By **working hypothesis 2.1** (here P_t is known and ζ_t independent), this is

$$\mathbb{E}_{H_1} \left[\left(\prod_{t=1}^{n-1} \exp(-\theta h(Y_t)) \right) \mathbb{E}_{1,P_n} [\exp(-\theta h(Y_n))] \right]$$

Note

$$\mathbb{E}_{1,P_n} [\exp(-\theta h(Y_n))] = \exp(\log \phi_{P_n,h}(\theta)) \leq \exp(\log \phi_{\mathcal{P},h}(\theta))$$

where $\phi_{\mathcal{P},h}(\theta) = \sup_{P \in \mathcal{P}} \phi_{P,h}(\theta) = \sup_{P \in \mathcal{P}} \mathbb{E}_{1,P} [\exp(-\theta h(Y))]$.

And by repeatedly doing above from $t = n - 1$ to $t = 1$,

$$\mathbb{E}_{H_1} \left[\prod_{t=1}^n \exp(-\theta h(Y_t)) \right] \leq (\exp(\log \phi_{\mathcal{P},h}(\theta)))^n = \exp(n \cdot \log \phi_{\mathcal{P},h}(\theta))$$

For the last equality in the proof,

$$\inf_{\theta \geq 0} \exp \left(\theta \mathbb{E}_0 h(Y) + \log \phi_{\mathcal{P},h}(\theta) \right) = \exp \left(\inf_{\theta \geq 0} [\theta \mathbb{E}_0 h(Y) + \log \phi_{\mathcal{P},h}(\theta)] \right)$$

as $\exp$ is continuous and monotone increasing.

Intuition for equation (29) and its proof

To control type 1 error, for $\alpha \in (0, 1)$

$$\mathbb{P}_{H_0} \left(\sum_{t=1}^n h(Y_t) \geq \gamma_{n,\alpha} \right) = \alpha$$

which is

$$\mathbb{P}_{H_0} \left(\frac{1}{n} \sum_{t=1}^n h(Y_t) \geq \frac{\gamma_{n,\alpha}}{n} \right) = \alpha$$

Under H_0 , Y_i are iid. Thus by SLLN,

$$\frac{1}{n} \sum_{t=1}^n h(Y_t) \xrightarrow{a.s.} \mathbb{E}_0 h(Y)$$

So we guess

$$\lim_{n \rightarrow \infty} \frac{\gamma_{n,\alpha}}{n} = \mathbb{E}_0 h(Y)$$

because otherwise, α will go to either 0 or 1.

Using contradiction, the paper shows

$$\limsup_{n \rightarrow \infty} \frac{\gamma_{n,\alpha}}{n} \leq \mathbb{E}_0 h(Y)$$

and

$$\liminf_{n \rightarrow \infty} \frac{\gamma_{n,\alpha}}{n} \geq \mathbb{E}_0 h(Y)$$

and so the limit exists and equals $\mathbb{E}_0 h(Y)$.

KL Divergence and Donsker-Varadhan

Definition 3.

Let P and Q be two probability measures on a measurable space $(\mathcal{X}, \mathcal{F})$. Assume P is absolutely continuous with respect to Q (denoted as $P \ll Q$), meaning that for any measurable set A , if $Q(A) = 0$, then $P(A) = 0$. By the Radon-Nikodym theorem, there exists a density function (or likelihood ratio) $\frac{dP}{dQ}$ such that $P(A) = \int_A \frac{dP}{dQ} dQ$.

The KL divergence from Q to P is defined as:

$$D_{\text{KL}}(P||Q) = \mathbb{E}_P \left[\log \frac{dP}{dQ} \right] = \int_{\mathcal{X}} \log \left(\frac{dP}{dQ} \right) dP$$

If P is not absolutely continuous with respect to Q , then $D_{\text{KL}}(P||Q) = \infty$.

Remark 4.

Note that every distribution is a probability measure (push forward measure):

$F_X(x) = \mathbb{P} \circ X^{-1}((-\infty, x])$. By Carathéodory extension thm, CDF uniquely determines its probability measure.

Remark 5.

By expectation Jensen's inequality, where $-\log$ is convex, $D_{\text{KL}}(P\|Q) \geq 0$, and $D_{\text{KL}}(P\|Q) = 0$ if and only if $P = Q$ almost everywhere.

Theorem 6 (Jensen's inequality).

Suppose that X is a random variable with expectation μ , and function g is convex and finite. Then

$$\mathbb{E}_X[g(X)] \geq g(\mathbb{E}_X[X])$$

with equality if and only if, for every line $a + bx$ that is a tangent to g at μ

$$\mathbb{P}_X[g(X) = a + bX] = 1.$$

<https://www.math.mcgill.ca/dstephens/556-2014/Handouts/Math556-05-Inequalities.pdf>

Corollary 7.

Suppose g is strictly convex and finite. Then $\mathbb{E}[g(X)] = g(\mathbb{E}[X])$ if and only if $\mathbb{P}[X = \mu] = 1$

Proof.

$\implies$: Assume $\mathbb{E}[g(X)] = g(\mathbb{E}[X])$. By theorem, for any tangent line $L(x) = a + bx$ to g at the point μ , we have $\mathbb{P}[g(X) = a + bX] = 1$. Because g is convex and finite, it possesses at least one subgradient at μ (This is fact 9.12 and corollary 9.13 in my co463 notes). Therefore, there exists at least one line $L(x) = a + bx$ that is tangent to g at μ . This guarantees the existence of a and b such that $g(\mu) = a + b\mu$. By the definition of strict convexity, a strictly convex function intersects any of its tangent lines at exactly one point: the point of tangency. Therefore, for all $x \neq \mu$, we have a strict inequality: $g(x) > a + bx$. Consequently, the equation $g(x) = a + bx$ holds true if and only if $x = \mu$. Define the events $A = \{\omega \in \Omega \mid g(X(\omega)) = a + bX(\omega)\}$ and $B = \{\omega \in \Omega \mid X(\omega) = \mu\}$. Then $g(X(\omega)) = a + bX(\omega) \iff X(\omega) = \mu$ holds for every ω by above. That is $A = B$. Recall we know $\mathbb{P}(A) = 1$. So $\mathbb{P}(B) = 1$, which means $\mathbb{P}[X = \mu] = 1$.

$\impliedby$: Assume $\mathbb{P}[X = \mu] = 1$. Let $L(x) = a + bx$ be an arbitrary tangent line to g at μ . 4. Because $L(x)$ is tangent to g at μ , then $g(\mu) = a + b\mu$. Since $\mathbb{P}[X = \mu] = 1$, then $\mathbb{P}[g(X) = g(\mu)] = 1$, $\mathbb{P}[a + bX = a + b\mu] = 1$. Since $g(\mu) = a + b\mu$, then $\mathbb{P}[g(X) = a + bX] = 1$.

This holds for every line $a + bx$ tangent to g at μ . Thus by theorem, $\mathbb{E}[g(X)] = g(\mathbb{E}[X])$. □

Theorem 8 (The Donsker-Varadhan Representation).

For any two probability measures P and Q on $\mathcal{X}$,

$$D_{\text{KL}}(P\|Q) = \sup_f (\mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f])$$

where the supremum is taken over all bounded measurable functions $f : \mathcal{X} \rightarrow \mathbb{R}$ (or any measurable function such that the expectations are well-defined).

Remark 9.

We say "taken over any measurable function such that the expectations are well-defined" is because, if we only require boundedness (which is the bare minimum if we want the representation to be well defined), the optimal function $f^* = \log\left(\frac{dP}{dQ}\right) + \log c$ below is usually not bounded.

Proof.

We prove it by proving

- $\mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f] \leq D_{\text{KL}}(P\|Q)$ for all f ,
- the equality is achieved for some specific f .

Assume $D_{\text{KL}}(P\|Q) < \infty$, and so $P \ll Q$. Let $p = \frac{dP}{dQ}$ be the Radon-Nikodym derivative.

Consider the quantity $\mathbb{E}_P[f] - D_{\text{KL}}(P\|Q)$. We can write this as a single integral with respect to P :

$$\mathbb{E}_P[f] - D_{\text{KL}}(P\|Q) = \int_{\mathcal{X}} f dP - \int_{\mathcal{X}} \log(p) dP = \int_{\mathcal{X}} \log(e^f) dP - \int_{\mathcal{X}} \log(p) dP = \int_{\mathcal{X}} \log\left(\frac{e^f}{p}\right) dP$$

$$\mathbb{E}_P[f] - D_{\text{KL}}(P\|Q) = \int_{\mathcal{X}} f dP - \int_{\mathcal{X}} \log(p) dP = \int_{\mathcal{X}} \log(e^f) dP - \int_{\mathcal{X}} \log(p) dP = \int_{\mathcal{X}} \log\left(\frac{e^f}{p}\right) dP$$

Since $\log$ is concave, by Jensen's inequality, $(\mathbb{E}[\log(Z)] \leq \log(\mathbb{E}[Z]))$:

$$\int_{\mathcal{X}} \log\left(\frac{e^f}{p}\right) dP \leq \log\left(\int_{\mathcal{X}} \frac{e^f}{p} dP\right)$$

By Radon-Nikodym, $dP = p dQ$,

$$\log\left(\int_{\mathcal{X}} \frac{e^f}{p} p dQ\right) = \log\left(\int_{\mathcal{X}} e^f dQ\right) = \log \mathbb{E}_Q[e^f]$$

Thus

$$\mathbb{E}_P[f] - D_{\text{KL}}(P||Q) \leq \log \mathbb{E}_Q[e^f]$$

Which is

$$\mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f] \leq D_{\text{KL}}(P||Q)$$

It is clear that if the equality in the Jensen's inequality is attained, then we are done. Note that the function $\phi(t) = -\log(t)$ is strictly convex and finite in $t > 0$ ($\log$ is strictly concave).

By the above corollary, the equality holds iff $Z = \frac{e^{f^*}}{p} = c$ almost surely, where c is the mean of Z and f^* is our optimal function. Then

$$\frac{e^{f^*}}{p} = c \implies e^{f^*} = cp \implies f^* = \log p + \log c = \log \left(\frac{dP}{dQ} \right) + \log c$$

Substitute this optimal function f^* back into the objective:

$$\begin{aligned} & \mathbb{E}_P[f^*] - \log \mathbb{E}_Q[e^{f^*}] \\ &= \mathbb{E}_P[\log(p) + \log(c)] - \log \mathbb{E}_Q[e^{\log(p) + \log(c)}] \\ &= \mathbb{E}_P[\log(p)] + \log(c) - \log(c \cdot \mathbb{E}_Q[p]) \\ &= D_{\text{KL}}(P||Q) + \log(c) - \log(c) - \log(1) \\ &= D_{\text{KL}}(P||Q) \end{aligned}$$

where

$$\mathbb{E}_Q[p] = 1$$

since $p dQ = dP$ and P is a probability measure, so $\int_{\mathcal{X}} p dQ = \int_{\mathcal{X}} dP = 1$.

Therefore the sup is attained by f^* .

Note that from the derivation above, such f^* is not unique, we can take any valid c we want.

We usually take $c = 1$.

□

Proof of Theorem 3.2, first part

Theorem 10.

Let $h_{\text{gum},\Delta}^* : [0, 1] \mapsto \mathbb{R}$ be defined as

$$h_{\text{gum},\Delta}^*(r) = \log \left(\left\lfloor \frac{1}{1-\Delta} \right\rfloor r^{\frac{\Delta}{1-\Delta}} + r^{\frac{\tilde{\Delta}}{1-\tilde{\Delta}}} \right) \quad \text{with} \quad \tilde{\Delta} = (1-\Delta) \left\lfloor \frac{1}{1-\Delta} \right\rfloor.$$

This function gives the optimal $\mathcal{P}_\Delta$ -efficiency rates in the sense that $R_{\mathcal{P}_\Delta}(h_{\text{gum},\Delta}^*) \geq R_{\mathcal{P}_\Delta}(h)$ for any measurable function h . Moreover, $R_{\mathcal{P}_\Delta}(h_{\text{gum},\Delta}^*)$ is attained at the following least-favorable NTP distribution in $\mathcal{P}_\Delta$:

$$P_\Delta^* = \left(\underbrace{1-\Delta, \dots, 1-\Delta}_{\left\lfloor \frac{1}{1-\Delta} \right\rfloor \text{ times}}, 1 - (1-\Delta) \cdot \left\lfloor \frac{1}{1-\Delta} \right\rfloor, 0, \dots \right)$$

The paper proves $R_{P_\Delta}(h_{\text{gum},\Delta}^*) \geq R_{P_\Delta}(h)$ by doing the following:

Recall from "optimality via minimax optimization", finding the optimal h that gives $\sup_h R_{\mathcal{P}}(h)$ is reduced to solving $\min_h \max_{P \in \mathcal{P}} L(h, P)$ where $L(h, P) := \mathbb{E}_0 h(Y) + \log \mathbb{E}_{1,P} e^{-h(Y)}$.

The paper shows

$$\min_h \max_{P \in \mathcal{P}} L(h, P) \geq -D_{KL}(\mu_0, \mu_1, P_\Delta^*)$$

But the paper also shows the min is attained at

$$\max_{P \in \mathcal{P}_\Delta} L(h_{\text{gum},\Delta}^*, P) = -D_{KL}(\mu_0, \mu_1, P_\Delta^*)$$

And so this $h_{\text{gum},\Delta}^*$ is the (*unique*) score function that solves the minimax problem

The following are the technical details:

We first claim that the radon-nikodym derivative $\frac{d\mu_{1,P_\Delta^*}}{d\mu_0}$ exists, that is, $\mu_{1,P_\Delta^*} \ll \mu_0$.

Proof.

It is clear that μ_0 , which is Y_t^{gum} under H_0 , follows a $U(0, 1)$ distribution. Hence μ_0 is the Lebesgue measure on $[0, 1]$. Thus for any mble set A , $\mu_0(A) = \int_{A \cap [0,1]} 1 dr = 0$ implies $A \cap [0, 1]$ has length 0.

μ_{1,P_Δ^*} , which is the distribution of Y_t^{gum} under H_1 at the "least-favorable NTP distribution". By lemma 3.1, we have CDF

$$\sum_{w \in \mathcal{W}} P_w r^{1/P_w}$$

for $r \in (0, 1)$. It is clear that since $U_{t,w} \sim U(0, 1)$, for $r \leq 0$ and $r \geq 1$, the CDF is 0 and 1 respectively. Thus we see the support of μ_{1,P_Δ^*} is between $[0, 1]$.

Now

$$\mu_{1,P_\Delta^*}(A) = \int_{A \cap [0,1]} f_1(r) dr$$

where $f_1(r)$ is the density of μ_{1,P_Δ^*} distribution (we will derive it later).

Therefore, if $\mu_0(A) = 0$, then $\mu_{1,P_\Delta^*}(A) = 0$, hence $\mu_{1,P_\Delta^*} \ll \mu_0$.

□

Thus we can define the radon-nikodym derivative

$$\frac{d\mu_{1,P_\Delta^*}}{d\mu_0}$$

Consider equation 21:

To apply the Donsker-Varadhan representation (we are allowed to do so because the radon-

nikodym derivative exists), use the substitution $f = -h$:

$$\begin{aligned}\min_h \left(\mathbb{E}_0[h(Y)] + \log \mathbb{E}_{1,P}[e^{-h(Y)}] \right) &= \min_f \left(\mathbb{E}_0[-f(Y)] + \log \mathbb{E}_{1,P}[e^{f(Y)}] \right) \\ &= -\max_f \left(\mathbb{E}_0[f(Y)] - \log \mathbb{E}_{1,P}[e^{f(Y)}] \right)\end{aligned}$$

By the Donsker-Varadhan representation (with $P = \mu_0$ and $Q = \mu_{1,P}$),

$$-\max_f \left(\mathbb{E}_{\mu_0}[f] - \log \mathbb{E}_{\mu_{1,P}}[e^f] \right) = -D_{\text{KL}}(\mu_0 \parallel \mu_{1,P})$$

We know what is this f , it is the f^* that we derived before! But we do not know what is $\log \left(\frac{dP}{dQ} \right)$.

By definition,

$$\begin{aligned}\mu_0(A) &= \int_A f_0(r) \, dr \\ \mu_{1,P_\Delta^*}(A) &= \int_A f_1(r) \, dr\end{aligned}$$

where f_0, f_1 are the density function of distributions μ_0 and μ_{1,P_Δ^*} .

We wish to find g such that

$$\mu_{1,P_\Delta^*}(A) = \int_A g(r) \, d\mu_0$$

From definition, it suffices to find

$$\mu_{1,P_\Delta^*}(A) = \int_A g(r) (f_0(r) \, dr)$$

Compare with above (because $f_0(r) = 1 \neq 0$),

$$g(r) = \frac{f_1(r)}{f_0(r)}$$

Therefore

$$\frac{d\mu_{1,P_\Delta^*}}{d\mu_0} = \frac{f_1(r)}{f_0(r)}$$

Now back to the proof of theorem 3.2, in equation 21,

As we have seen in the proof of The Donsker-Varadhan Representation, the specific $h = -f$ that makes the equality hold (which is also our desired minimizer h) is (by taking $c = 1$, so $\log c = 0$)

$$h_{\text{gum},\Delta}^*(r) = \log \frac{d\mu_{1,P_\Delta^*}}{d\mu_0}(r) = \log \left(\frac{f_1(r)}{f_0(r)} \right)$$

Now we need to figure out what that is. It is clear that $f_0(r) = 1$

What is $f_1(r)$?

By lemma 3.1, under H_1 , the distribution of Y_t^{gum} given P_t obeys

$$\mathbb{P}_{H_1}(Y_t^{gum} \leq r \mid P_t) = \sum_{w \in \mathcal{W}} P_{t,w} r^{1/P_{t,w}} \quad \text{for } r \in [0, 1]$$

By taking its derivative, we obtain the density:

$$\frac{d}{dr} \left(\sum_{w \in \mathcal{W}} P_{t,w} r^{1/P_{t,w}} \right) = \sum_{w \in \mathcal{W}} P_w \left(\frac{1}{P_w} r^{\frac{1}{P_w}-1} \right) = \sum_{w \in \mathcal{W}} r^{\frac{1-P_w}{P_w}}$$

By subbing in P_Δ^* , we can get $f_1(r)$.

Recall

$$\begin{aligned} P_\Delta^* &= \left(\underbrace{1 - \Delta, \dots, 1 - \Delta}_{\lfloor \frac{1}{1-\Delta} \rfloor \text{ times}}, 1 - (1 - \Delta) \cdot \left\lfloor \frac{1}{1 - \Delta} \right\rfloor, 0, \dots \right) \\ &= \left(\underbrace{1 - \Delta, \dots, 1 - \Delta}_{\lfloor \frac{1}{1-\Delta} \rfloor \text{ times}}, 1 - \tilde{\Delta}, 0, \dots \right) \end{aligned}$$

by defining $\tilde{\Delta} = (1 - \Delta) \lfloor \frac{1}{1-\Delta} \rfloor$.

Therefore

$$f_1(r) = \left\lfloor \frac{1}{1 - \Delta} \right\rfloor r^{\frac{\Delta}{1-\Delta}} + r^{\frac{\tilde{\Delta}}{1-\Delta}}$$

and we get the desired $h_{gum,\Delta}^*$, which is non-decreasing in r .

And the rest is clear by the paper.

□

Proof of lemma 3.2, 3.3 and lemma 3.4

lemma 3.2:

derivative: Use $r^{1/P_i} = e^{\frac{\ln r}{P_i}}$

lemma 3.3:

Remark 11.

$\mathcal{P}_\Delta$ is compact and convex by definition

Proof.

By definition $\mathcal{P}_\Delta := \{\mathbf{P} : \max_{w \in \mathcal{W}} P_w \leq 1 - \Delta\}$, which is

$$\mathcal{P}_\Delta = \left\{ \mathbf{P} \in \mathbb{R}^{|\mathcal{W}|} : \sum_{w \in \mathcal{W}} P_w = 1, \quad 0 \leq P_w \leq 1 - \Delta \quad \text{for all } w \in \mathcal{W} \right\}$$

Verification of convexity is clear.

To prove compactness, since we are in $\mathbb{R}^n$, by Heine-Borel, it suffices to prove closed and bounded.

Closeness:

- Let $A := \{\mathbf{P} \in \mathbb{R}^{|\mathcal{W}|} : \sum_{w \in \mathcal{W}} P_w = 1\}$. The function $f(\mathbf{P}) = \sum P_w$ is continuous. The set A is the preimage of the closed singleton set $\{1\}$ under f , meaning A is closed.
- For each $w \in \mathcal{W}$, $B_w = \{\mathbf{P} \in \mathbb{R}^{|\mathcal{W}|} : P_w \geq 0\}$. The projection function $g_w(\mathbf{P}) = P_w$ is continuous. The set B_w is the preimage of the closed interval $[0, \infty)$ under g_w , meaning B_w is closed.
- For each $w \in \mathcal{W}$, $C_w = \{\mathbf{P} \in \mathbb{R}^{|\mathcal{W}|} : P_w \leq 1 - \Delta\}$. Using the same continuous projection $g_w(\mathbf{P}) = P_w$, C_w is the preimage of the closed interval $(-\infty, 1 - \Delta]$, meaning C_w is closed.

Note

$$\mathcal{P}_\Delta = A \cap \left(\bigcap_{w \in \mathcal{W}} B_w \right) \cap \left(\bigcap_{w \in \mathcal{W}} C_w \right)$$

Thus it is closed.

Boundedness:

$$\|\mathbf{P}\|_2 = \sqrt{\sum_{w \in \mathcal{W}} P_w^2} \leq \sqrt{\sum_{w \in \mathcal{W}} 1^2} = \sqrt{|\mathcal{W}|}$$

lemma 3.4:

□

What does it mean by **permutation invariance**:

$$\phi_h(\mathbf{P}) = \int_0^1 e^{-h(r)} dF_{1,P}(r)$$

where $F_{1,P}(r) = \sum_{w=1}^{|\mathcal{W}|} P_w r^{1/P_w}$.

Proof of lemma 3.4:

What does it mean by "It suffices to demonstrate that any point in $\mathcal{P}_\Delta$ is in the convex hull of $\Pi(\mathbf{P}_\Delta^*)$, denoted by $\text{Conv}(\Pi(\mathbf{P}_\Delta^*))$ ":

We wish to prove

$$\mathcal{P}_\Delta = \text{Conv}(\Pi(\mathbf{P}_\Delta^*))$$

because of the following:

Remark 12.

If a convex set is generated by the convex hull of a set of points , then the extreme points of that convex set must necessarily belong to this set of points .

“It is easy to find that P is the convex interpolation of $|\mathcal{I}|$ points in

$\Pi(P_{\Delta}^*)$ whose first N coordinates are the same as $1 - \Delta$

and the i th coordinate is $1 - N(1 - \Delta)$ ”：对于集合 $\mathcal{I}$ 中的每一个索引 i ，我们在极值点集合 $\Pi(P_{\Delta}^*)$ 中选取一个特定的极值点 i_0 。这个极值点 i_0 ：前 m 个坐标等于 $1 - \Delta$ ，第 i_0 个坐标等于剩余概率 R ，其余所有坐标等于 0。为每个 i 分配一个权重 $w_i = \frac{P_i}{R}$ 。对于前 m 个坐标：

$\sum_{i \in \mathcal{I}} w_i (1 - \Delta) = (1 - \Delta) \sum w_i = 1 - \Delta$ 。与 P 一致。对于第 $j \in \mathcal{I}$ 个坐标：在所有选取的极值点中，只有 j_0 在这个位置非零。因此其值为 $w_{j_0} \cdot R = \frac{P_{j_0}}{R} \cdot R = P_{j_0}$ 。与 P 一致。对于其他坐标：恒为 0，与 P 一致。

In (a), for

$$P = \theta P_1 + (1 - \theta) P_2$$

need a valid θ . On m -th coordinates,

$$P_m = \theta(1 - \Delta) + (1 - \theta)(P_m + P_{m+1} - (1 - \Delta))$$

So

$$\theta = \frac{(1 - \Delta) - P_{m+1}}{((1 - \Delta) - P_m) + ((1 - \Delta) - P_{m+1})}$$

which is in $(0,1)$ as $1 - \Delta > P_m \geq P_{m+1}$. Sub in θ to $m + 1$ coordinates we also get P_{m+1} .